

GINGIN, Australia

WMO: 956120

Lat: 31.463S

Lon: 115.864E

Elev: 75

StdP: 100.43

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.5	2.9	-3.2	2.9	19.8	-1.2	3.4	16.9	10.3	15.5	9.4	15.1	1.2	130	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.1	37.6	19.4	35.6	19.1	33.8	18.9	22.3	30.1	21.4	29.5	20.6	28.7	5.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	15.1	24.1	19.2	14.1	23.4	18.2	13.3	22.9	65.9	30.2	62.4	29.4	59.8	28.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.2	8.5	-1.1	41.5	1.2	1.7	-2.0	42.7	-2.6	43.6	-3.3	44.6	-4.2	45.8		
(5)			-1.3	23.9	1.4	1.0	-2.4	24.6	-3.2	25.2	-4.0	25.7	-5.1	26.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	24.3	24.4	22.7	19.0	15.6	13.1	11.9	12.6	13.6	16.2	19.8	21.8
	DBStd	5.49	3.60	3.58	3.77	3.37	2.88	2.41	2.43	2.29	2.45	3.18	3.74	3.95
	HDD10.0	18	0	0	0	0	1	3	10	3	1	0	0	0
	HDD18.3	932	1	2	6	31	94	158	199	177	145	83	26	9
	CDD10.0	2895	444	403	394	271	173	96	69	85	108	193	294	365
	CDD18.3	766	187	172	142	52	8	1	0	0	2	18	69	116
	CDH23.3	8444	2079	1878	1460	481	96	3	0	7	43	237	812	1349
CDH26.7	3991	1089	947	672	155	16	0	0	1	7	72	359	672	
Wind	WSAvg	4.2	5.1	5.0	4.8	3.7	3.3	3.4	3.3	3.6	3.8	4.2	4.7	5.0
Precipitation	PrecAvg	669	16	14	20	33	88	122	132	108	68	36	21	11
	PrecMax	869	101	80	86	74	167	234	205	166	138	88	63	46
	PrecMin	442	0	0	0	6	43	47	55	23	14	12	2	0
	PrecStd	91	28	17	20	19	35	46	38	32	29	18	18	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	39.2	38.4	33.4	28.5	23.5	21.6	24.1	27.3	32.3	36.7	39.4
		MCWB	19.2	20.3	19.6	17.5	15.3	13.3	13.6	14.1	15.2	17.1	17.4	18.6
	2%	DB	37.1	36.6	35.3	30.4	26.0	21.6	19.7	21.6	24.1	28.4	33.6	36.3
		MCWB	19.4	19.7	19.4	17.5	15.1	13.9	13.4	14.2	14.7	16.3	17.3	18.5
	5%	DB	34.9	34.5	32.7	28.3	23.8	20.0	18.3	19.6	21.6	25.7	30.8	33.3
		MCWB	19.4	19.4	18.9	17.2	14.7	13.5	13.3	13.5	14.2	15.9	17.0	18.3
	10%	DB	32.6	32.4	30.4	26.1	21.9	18.6	17.1	18.1	19.6	23.4	28.0	30.4
		MCWB	19.1	19.4	18.5	16.9	14.7	13.2	12.8	13.3	13.7	15.4	16.7	17.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.2	23.7	23.2	20.8	19.3	17.6	16.9	17.4	17.5	19.1	20.5	21.4
		MCDB	30.8	32.1	30.4	26.1	22.6	19.8	18.2	19.7	22.2	25.9	28.9	33.0
	2%	WB	22.1	22.4	21.7	19.8	17.9	16.3	15.3	16.0	16.3	17.9	19.4	20.3
		MCDB	29.9	30.5	28.9	25.2	21.1	18.7	17.4	18.5	20.9	25.3	27.9	30.8
	5%	WB	21.2	21.2	20.7	19.0	16.9	15.3	14.3	15.0	15.3	17.0	18.6	19.5
		MCDB	29.2	29.9	27.6	24.4	20.7	17.8	16.6	17.9	19.5	23.1	26.5	28.8
	10%	WB	20.3	20.4	19.8	18.1	15.9	14.2	13.5	14.0	14.5	16.1	17.7	18.7
		MCDB	28.6	29.0	26.8	23.8	19.9	17.0	15.9	16.9	18.4	21.9	25.0	27.0
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.1	13.7	13.3	12.8	11.4	11.0	11.3	12.2	13.6	14.2	14.3
		MCDBR	18.0	16.6	16.4	15.9	15.8	14.0	12.4	14.5	16.6	17.0	18.3	18.8
		MCWBR	5.5	5.2	5.6	6.4	7.6	7.9	7.5	8.2	8.6	7.3	6.3	5.9
	5% WB	MCDBR	14.0	14.2	12.6	12.4	11.2	10.1	9.3	10.4	13.1	14.3	15.3	15.5
		MCWBR	5.1	5.3	5.4	6.1	6.9	7.1	6.6	6.8	7.6	6.9	6.2	5.9
Clear-Sky Solar Irradiance	taub	0.340	0.341	0.325	0.316	0.306	0.296	0.297	0.303	0.324	0.325	0.326	0.328	
	taud	2.563	2.573	2.609	2.620	2.617	2.631	2.620	2.601	2.517	2.533	2.555	2.568	
	Ebn at Noon	1000	979	963	920	881	867	882	921	947	984	1007	1014	
	Edh at Noon	108	103	93	83	74	69	73	83	100	106	108	108	
All-Sky Solar Radiation	RadAvg	8.23	7.44	6.18	4.57	3.44	2.83	2.99	3.86	5.13	6.63	7.84	8.44	
	RadStd	0.37	0.24	0.30	0.25	0.23	0.20	0.18	0.17	0.28	0.28	0.29	0.34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page